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Palaeomagnetism and palaeogeography of the Southern Uplands block in Ordovician times

J. D. A. Piper
Sub-department of Geophysics, University of Liverpool, Liverpool L69 3BX

Synopsis

The major Ordovician igneous outcrops within the Southern Uplands were studied to evaluate the position of this region with respect to adjacent crustal plates during the later history of the Iapetus Ocean. Post-Arenig, pre-Caradocian gabbros and serpentinites of the Ballantrae Complex yield palaeomagnetic poles of 26°E, 10°S ($d\psi$ = 10·9, $d\chi$ = 18·8°) and 27°E, 12°S ($d\psi$ = 5.5, $d\chi$ = 9·7°) respectively after a.f. and thermal cleaning. Although 95 per cent confidence circles overlap with other British Ordovician data, magnetic inclinations are systematically shallower than contemporaneous results from England and Wales and correlate closely with other data from N of the Southern Uplands Fault line. A similar result is found from palaeomagnetic study of the (Lower Caradocian) Bail Hill Volcanics. The palaeomagnetic results support faunal and tectonic evidence that the major crustal separation in Ordovician times existed between the Southern Uplands and the northern Lake District. The visible ophiolite slices along the approximate line of the present Midland Valley represent pre-Llanvirnian obduction in a northerly direction of an earlier ocean basin.

Introduction

The Caledonian orogenic cycle was associated with the development and consumption of the Iapetus (Proto-Atlantic) ocean basin (Wilson 1966, Harland and Gayer 1972). The Ballantrae complex is an obducted ophiolite relict of this ocean (Dewey 1969), S of which a trench was infilled with turbidites and flysch-facies sediments in Ordovician and Silurian times. The relative position of this region is critical to models linking the destruction of the basin to stages in the development of the orthotectonic Caledonides to the N and the paratectonic Caledonides to the S. The documented models include an oceanic plate thrust northwards under the orthotectonic Caledonides (Dewey and Pankhurst 1970), a plate thrust southwards under the broad area of the paratectonic Caledonides (Fitton and Hughes 1970), paired subduction zones beneath the Highland Boundary Fault and Southern Uplands Fault with the Midland Valley as a relict of the Iapetus Ocean (Gunn 1973), northwesterly subduction beneath the Southern Uplands (Mitchell and McKerrow 1975), and convergence between the Southern Uplands region and Northern England along a triple junction migrating progressively to the SW (Phillips *et al.* 1976).

Exposed evidence for the development of the Southern Uplands region dates

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largely from post-Arenig times, and the possibilities of resolving contradictions between these models by palaeomagnetic studies (which may determine the palaeolatitude but not the palaeolongitude of the regions concerned) are severely restricted by the limited degree of separation evident at this time. Briden *et al.* (1973, fig. 3) find that Lower Ordovician rock units from the British Isles have systematically shallower palaeomagnetic inclinations in regions N of the Highland Boundary Fault compared with regions S of the fault. Better consistency with the geomagnetic axial dipole model is achieved if a crustal separation of about 1000 km is postulated between the Highland Boundary Fault and the Northern Lake District. Unfortunately, because of the statistical nature of palaeomagnetic results this estimate has a large associated uncertainty (±800 km), and it is only because *all* results from S of the postulated suture have steeper palaeomagnetic inclinations to *all* results from N of the suture that the observation can be viewed with some confidence.

Development of the Southern Uplands Block thus took place within a narrow ocean basin, and defining its position within this basin by palaeomagnetic study is further hampered by the paucity of relatively undeformed igneous rocks. The exceptions are certain units of the Ballantrae Complex, Ayrshire, and the Bail Hill Volcanics of Dumfriesshire. The palaeomagnetism of these rocks is reported here.

Geological Outline

The Ballantrae Complex has formed the subject of a long history of geological study, and following the classical work of Nicol, Murchison, Lapworth, Peach and Horne, interest has continued in the present century with the structural and petrologic studies of Balsillie (1932), Anderson (1936) Bailey and McCallien (1957) and Williams (1962). During the past decade interest in the complex has centred on its ophiolitic affinities and, as the largest exposed body of its kind in the British Caledonides, is critical to the models listed above. The rocks available for study are hardly suitable for a complete resolution of this problem because the Middle Arenig pillow lavas interpreted as ancient Iapetus crust (Dewey 1971) were highly deformed and subjected to high pressure, low temperature metamorphism. Deformation was accompanied by widespread development of serpentinite and intrusion of gabbro; these rocks provide opportunity to study later events. Sampling for this study has accordingly been concentrated on the Byne Hill gabbro together with the main serpentinite outcrop where it is exposed to advantage in coastal sections (Fig. 1).

A stage represented by spilitic pillow lavas, shales, cherts and volcanogenic sediments containing Arenig (*Didymograptus protobifidus* Zone) graptolites was highly deformed by ophiolite obduction with green-schist to blue-schist facies metamorphism, prior to eruption of a younger suite of spilitic lavas of possible Llandeilo age (Whittington 1972) now exhibiting mild deformation (Mendum 1968). The igneous associations at Ballantrae are not disposed in the classic ophiolitic sequence but all components of such a sequence are present. Bailey and McCallien (1957) regarded the

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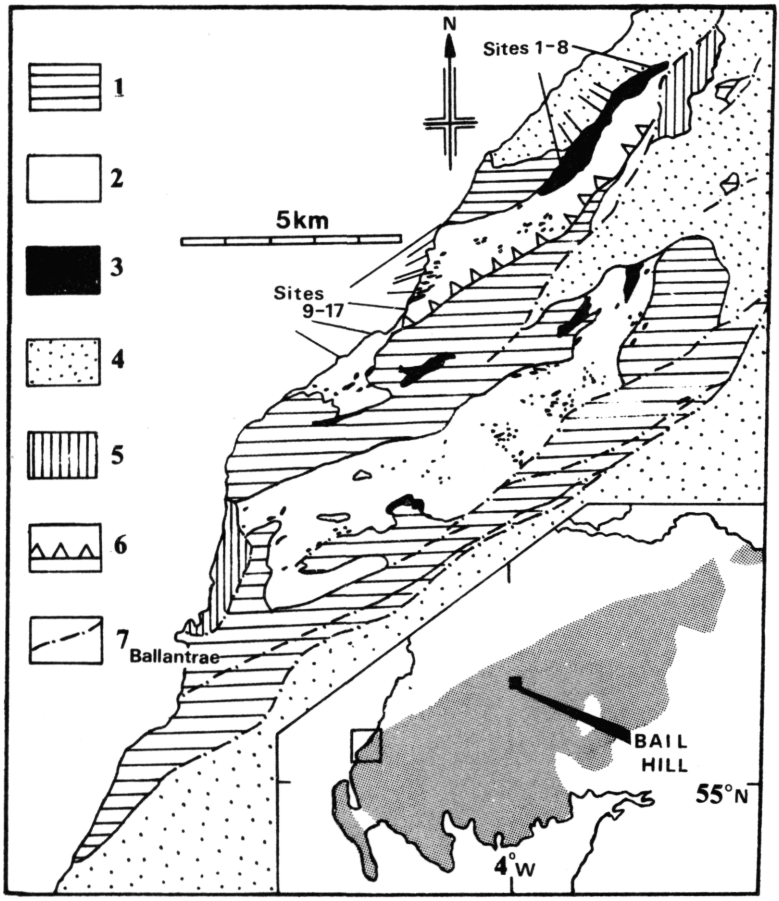


FIG. 1. Simplified Geological map and sampling sites of the Ballantrae Complex, and (inset) location of rock units sampled for this study within the Southern Uplands palaeozoic outcrop. Key to symbols: 1, pillow lavas and volcanogenic sediments; 2, serpentinites; 3, gabbros; 4, post-Ballantrae Complex, Lower Palaeozoic rocks; 5, Upper Palaeozoic rocks; 6, thrust; 7, faults.

serpentinite (Fig. 1) as a single sheet arching over the Bennane Head "anticline" and separating two volcanic series but later interpretations (e.g. Dewey 1971) consider the volcanics to belong to the single uppermost unit of an ophiolite succession. Peach and Horne (1899) regarded the serpentinites (palaeomagnetic sites 9 to 17 of this study, Fig. 1) as intrusive into the lava cover while Balsillie and Pringle regarded it as Precambrian basement to the volcanic sequence. More detailed studies of contact relations demonstrate that the serpentinite developed from metasomatic alteration of ultramafic and volcanic assemblages (Anderson 1936, Bailey and McCallien 1957). The hydrothermal process of rodingitisation has subsequently widely and sporadically affected the serpentinite (Balsillie 1932) and gabbro dykes intrusive into the

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serpentinite (Bloxam 1968). Gabbros, from massive to foliated in character, are present as many small intrusions variably altered to lower temperature assemblages and sometimes with inclusions of serpentine; the largest gabbro complex at Byne Hill (palaeomagnetic sites 1–8, Fig. 1) includes a complete differentation sequence from high-Na gabbro to biotite trondjemite.

Harris *et al.* (1965) dated samples of gabbro within the serpentinite outcrop 2·5 km N of Colmonell by the K–Ar method; they regarded the age 475 m.y. from five determinations (483, 467, 476, 477 and 479 m.y.) as dating the episode of intrusion and serpentinization (post-Middle Arenig and pre-Llandeilian). The complex is unconformably overlain by conglomerates succeeded by sediments containing a Lower Caradocian (Glenkiln) fauna.

The Bail Hill volcanic succession (palaeomagnetic sites 18–22) comprises a pile of andesitic lavas and pyroclastic rocks involved in the regional folding (Peach and Horne 1899) and overlying Arenig cherts. The volcanics are interstratified with shales containing Lower Caradocian graptolites and Pringle (1948) assigns all the Bail Hill volcanic succession to the Glenkiln horizon. Harris *et al.* (1965) report four K–Ar ages (455, 455, 440 and 437 m.y.) from mica andesite of the Bail Hill succession and assign the mean age of 445 m.y. as a minimum age to the Lower Caradocian. This compares with other age determinations of *c* 447 for the Lower Caradocian (Lambert 1971) and 451 m.y. for the mid-Caradoc (Actonian) beds of the Welsh Borderlands (Ross *et al.* 1976). The Bail Hill succession is thus about 30 m.y. younger than the Ballantrae Complex. The five palaeomagnetic sites were sampled through approximately 150 metres of the sequence separated by shale horizons and exposed in gullies on the west side of Bail Hill approximately 1·5 km NE of Kirkconnel.

Palaeomagnetic Results

Cores drilled in the field with a motorized portable drill were oriented by sun and/or magnetic sights and cut into 2·5 cm cylinders for palaeomagnetic measurements with a parastatic magnetometer. All samples were progressively demagnetized using an alternating field system in steps of 50 or 100 Oersteds (oe.) and remeasured after each step. The stability of magnetic vectors isolated by a.f. cleaning was tested by thermally-demagnetizing pilot samples in steps of 100°C to 500°C, then at 530°C and 560°C. Sections from the base of some cores were further utilized for polished section study, and thermomagnetic determinations (saturation magnetization, Js, versus temperature) were made on small fragments using a Chevallier balance.

Samples of both the serpentinites and gabbros exhibit a systematic fall in moments with progressive cleaning typical, but not diagnostic of titanomagnetite-held remanence, and directions stabilize with 10–50% of initial moments remaining (Fig. 2). These residual moments are progressively reduced by thermal demagnetization (Fig. 3) with no further systematic change in direction of magnetization; the demag-

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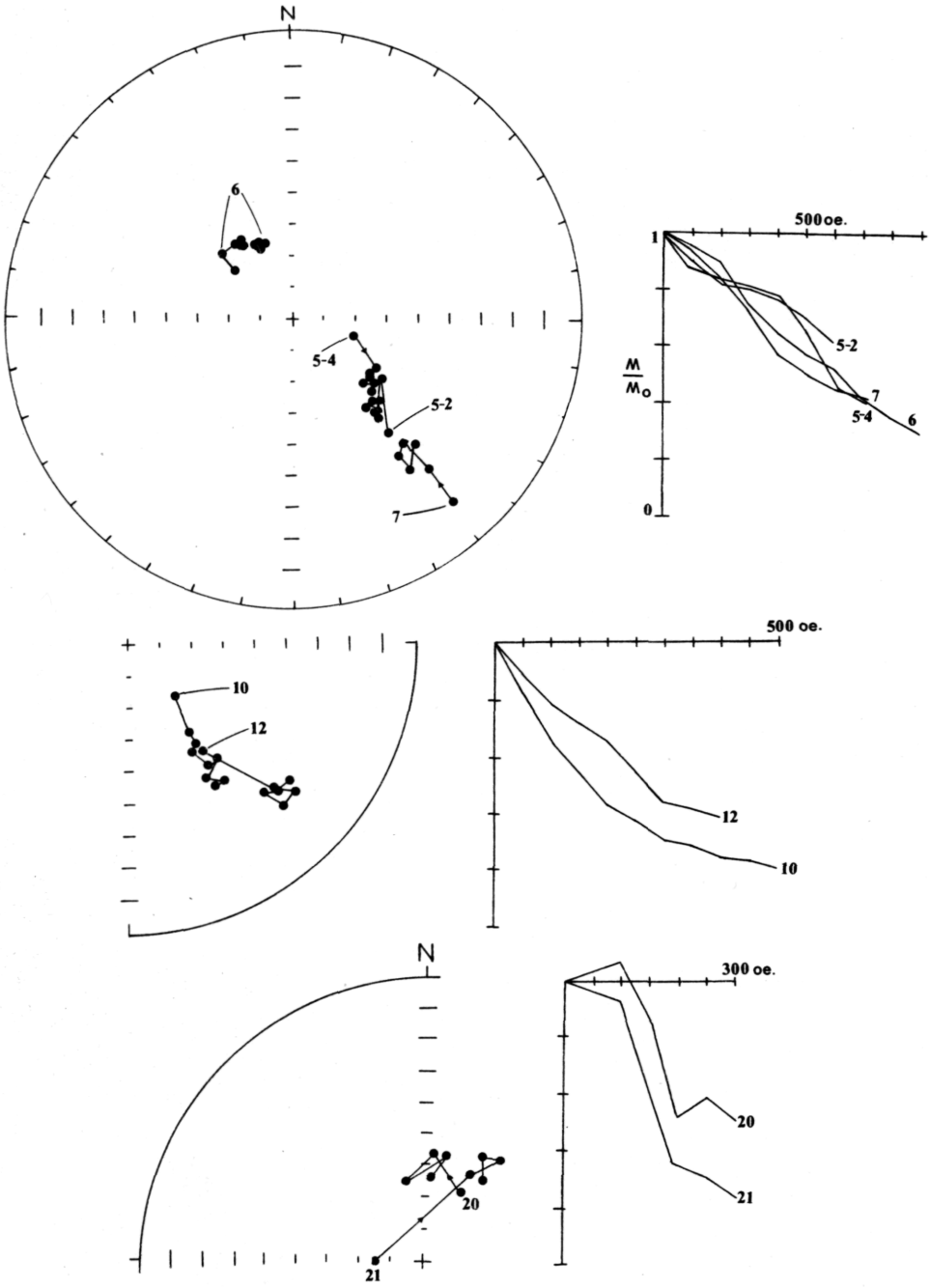


FIG. 2. Behaviour of *in situ* directions of magnetization and magnetic moments with progressive a.f. demagnetization. Moments are expressed as a fraction of the initial moment, Mo; projections are stereographic and open symbols are upper hemisphere plots and closed symbols are lower hemisphere plots. (Numbers refer to the site numbers of Tables I and II and Fig. 1.)

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netization spectra suggest that magnetic moments disappear at 500°C or slightly higher, an observation confirmed by thermomagnetic determinations (Fig. 5). After cleaning, all but three sites have similar directions of magnetization of one polarity, and cleaned site mean directions are listed in Table I. Sites 6, 10 and 13 have northerly positive directions of magnetization stable to a.f. and thermal cleaning (Figs. 2 and 3) and are excluded from the calculated group mean directions (Table I); these means show that the serpentinites and gabbros have closely similar directions of magnetization (D = 147·7, I = 32·4° (a_{95} = 8·6°) and D = 147·6°, I = 36·2° (a_{95} = 16·1°) respectively).

Late Silurian (Ludlow or Downtonian)tectonism involving thrusting and three stages of folding affected the cover rocks of the Ballantrae Complex (Williams 1962). The complex locally diverted the direction of principal stress and the effects of this deformation within pre-Caradoc infrastructure are difficult to resolve, although the close agreement of directions of magnetization between the serpentinites and gabbros implies either that the magnetization postdates these Caledonian movements or that these later movements did not effect major refolding within the complex. The former possibility is effectively ruled out by the agreement of the palaeomagnetic poles with

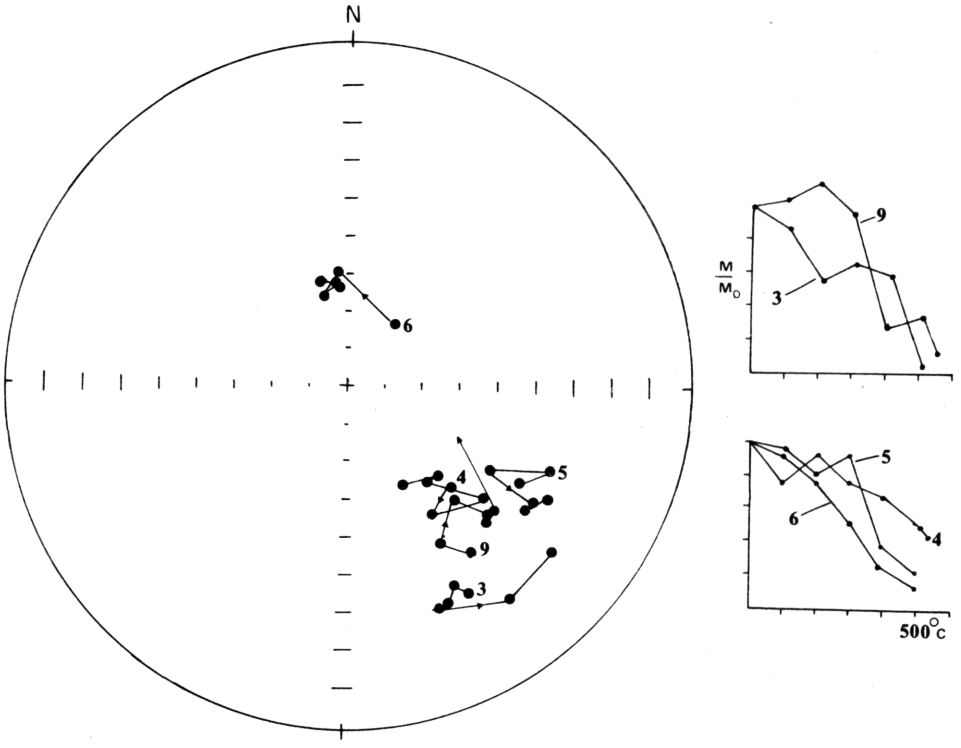


FIG. 3. Behaviour of directions of magnetization and magnetic moments with progressive thermal demagnetization; nomenclature is the same as for Figure 2.

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other British Ordovician data (summarized by Briden *et al.* 1973), although it is probable that the late Caledonian deformation has affected local directions of magnetization and influenced the overall grouping of site mean directions; a test for this cannot be made from the present data.

The other possible influence on the directions of magnetization at Ballantrae is the post-Caledonian regional tilting which Williams (1958) documents as a gentle eastward tilt affecting the district before the (pre-Lower Old Red Sandstone) wrench faulting. The highly deformed state of the cover rocks makes the magnitude of this tilt difficult to estimate but it is certainly slight, and being nearly at right angles to the directions of magnetization, can have had no appreciable effect on the palaeomagnetic directions.

TABLE 1
Palaeomagnetic results, Ballantrae gabbros and serpentinites, after a.f. and thermal cleaning

	N	D	I	R	k	α_{95}
(i) Gabbros						
1	5	146	+30	4·74	16	19·9
2	4	154	+14	3·77	13	26·5
3	5	140	+13	4·74	15	20·3
4	5	152	+67	4·76	17	19·3
5	7	142	+58	6·82	34	10·6
*6	4	338	+66	3·94	50	13·1
7	5	151	+32	4·21	5	37·8
8	6	149	+40	5·88	41	10·6
(ii) Serpentinites						
9	7	157	20	6·62	16	15·7
*10	7	24	78	6·85	41	9·5
11	6	153	35	5·92	65	8·4
12	7	148	32	6·75	24	12·5
*13	7	301	40	5·63	4	32·6
14	5	141	42	4·34	6	33·7
15	7	157	15	6·75	24	12·6
16	7	135	38	6·18	7	23·9
17	4	140	33	3·83	18	22·4
(iii) Combined mean directions and palaeomagnetic poles						
Gabbros	7	147·6	36·2	6·60	15	16·1
				Palaeomagnetic Pole 25·8°E, 9·9°S ($d\psi$ = 10·9°, $d\chi$ = 18·8°)		
Serpentinites	7	147·7	32·4	6·88	51	8·6
				Palaeomagnetic Pole 26·5°E, 12·3°S ($d\psi$ = 5·5°, $d\chi$ = 9·7°)		

* Sites excluded from group mean analysis.

Footnote: D and I are the mean declination and inclination of N sites or samples giving a vector resultant R; k is Fisher's precision parameter and equal to (N–1)/(N–R), and α_{95} is the radius of the 95 per cent confidence circle about the mean direction; $d\psi$ and $d\chi$ are the semi axes of the 95 per cent confidence oval about the palaeomagnetic pole in the colatitude direction and at right angles to it respectively.

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Nesbitt (1967) has earlier reported palaeomagnetic results from 12 samples in lavas at Stockenray Bay and after cleaning in 85 oe. found a better grouping prior to tilt correction. The mean direction (D = 189°, I = 40°, R = 11·5) is not widely removed from the overall mean directions of this study and since the tilt corrected direction (D = 202°, I = 21°, R = 9·2) is further from this mean, the results support her conclusion that these rocks were remagnetized during folding and metamorphism. Although sampled during the present study, the pillow lavas did not yield an overall significant mean direction of magnetization probably because of variable amounts of remagnetization during folding and metamorphism; it is considered that only detailed thermal demagnetization might be capable of resolving primary components in the Arenig lavas to produce a magnetic inclination relative to bedding (and hence a palaeolatitude) for these rocks.

The Bail Hill volcanics are stable to a.f. demagnetization and typically exhibit progressive fall in moment with treatment, although after demagnetizing to 200–300 oe. samples generally became too weak to measure and the site mean directions of Table II are accordingly given for demagnetization in fields of 150–250 oe. After

TABLE 2

Palaeomagnetic Results, Bail Hill Volcanics (356·1°E, 55·4°N), a.f. cleaned and dip corrected

Site No.	N	D	I	R	k	Intensity × 10^{-4} e.m.u./gm × 10^{-5}	Mass Susceptibility c.g.s. units
18	5	358	−2	4·76	17	0·19–0·03	0·31–0·17
19	6	4	−2	5·86	36	0·53–0·05	0·23–0·17
20	4	2	3	3·95	61	0·04–0·01	0·57–0·44
21	5	27	−10	4·78	18	0·48–0·10	0·95–0·34
22	12	6	−7	10·69	18	0·19–0·02	0·71–0·54
						Palaeomagnetic Pole	
		D	I	R	k	a_{95}	°E °N
Mean of 5 sites		7·3	−3·5	4·90	42	11·9°	167·5 32·5
						($d\psi$ = 6·0°, $d\chi$ = 12·0°)	

correction for tectonic dip of interstratified shales (40°–60° in directions between 352° and 4° E) the group mean direction is D = 7·3° I = −3·5° (a_{95} = 11·9°) with a palaeomagnetic pole at 167·5°E, 32·5°S and the overall precision (k) improves slightly from k = 22 to k = 42. Since the direction of magnetization prior to dip correction is slightly shallower than the present earth's field it is conceivable that the a.f. treatment has not completely removed components of viscous magnetization (VRM) although this is unlikely since directions of magnetization do not move systematically away from the present dipole field (Fig. 2) with progressive a.f. treatment. The biggest uncertainty in the derived pole is caused by the large tilt correction; this has been made here assuming simple concentric folding.

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Magnetic Properties

The lavas have a broad distribution of magnetic properties (Fig. 4) with no well defined Gaussian distribution, and although on the whole the plutonic rocks are slightly less magnetic than the volcanics, serpentinized rocks have a distinctly bimodal distribution. While the majority of the serpentinites are weak and comparable to their probable source rocks, small areas of the serpentinite outcrop exhibit very high magnetic intensities and susceptibilities; the processes of serpentinization and/or rodingitisation have clearly enhanced the magnetic mineral content of these rocks. Differences in the measured properties qualitatively reflect the abundances of magnetic constituents observed in polished sections. The pillow lavas typically comprise sulphide grains (mostly $<5\mu$) with titanomagnetite grains an order or more larger in

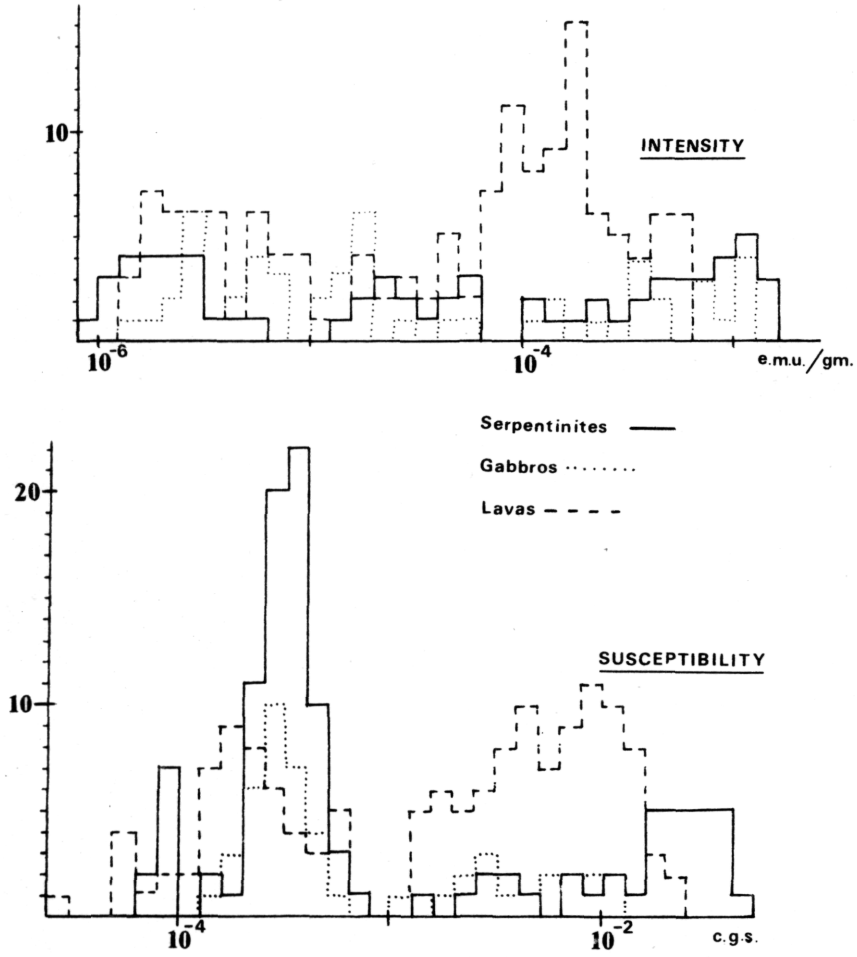


FIG. 4. Histogram of Magnetic intensities (prior to a.f. demagnetization) and susceptibilities of the different rock types within the Ballantrae Complex plotted on a logarithmic scale.

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size. Although deuteric (*c* 500–800°C) oxidation is minimal and equivalent to oxidation classes 1 and 2 of Ade-Hall *et al.* (1968), the common influence of hydrothermal alteration is evident from the oxidation of maghemite to hematite and variable alteration of ilmenite lamellae to rutile/hematite aggregates. Thermomagnetic determinations (Fig. 5) show that titanomagnetite near to pure magnetite in composition ($Tc \simeq 575°C$) is the principal remanence carrier, and 75% of the determinations exhibit a small kink in the curves at 160–200°C (Fig. 5) which probably results from unmixing of titanomagnetite and maghemite, a feature common in submarine lavas (Ozima and Ozima 1971). Within the gabbros maghemite rims are well developed around large (up to 1 mm) mostly unoxidised grains of titanomagnetite, and these samples also exhibit a kink in thermomagnetic curves over the same range. These minerals are

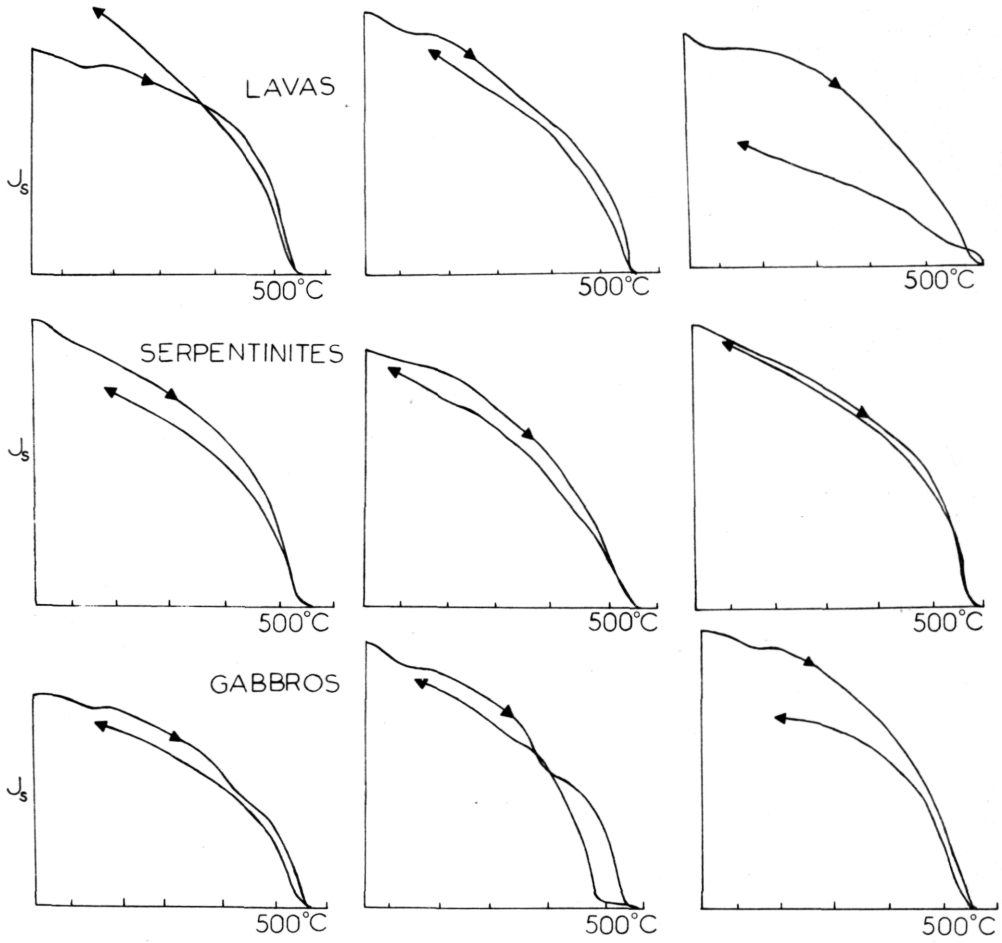


FIG. 5. Thermomagnetic determinations (saturation magnetization versus temperature) for some typical samples of pillow lavas, serpentinites and gabbros from the Ballantrae Complex.

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associated with large sulphide grains in the gabbros; effects of low temperature alteration are much less than in the Arenig lavas and most evident in grains with ilmenite lamellae.

Properties of the serpentinites are more difficult to systematize. Critical to this facies is the variable development of new magnetic phases reflected in the distribution of magnetic properties (Fig. 4) and Balsillie (1932) describes deposition of ilmenite and titanomagnetite along cleavage planes in ferromagnetic constituents; these grains do not exhibit characteristics of deuteric alteration. By contrast advanced stages of rodingitisation are represented by abundant hematite in highly reddened samples as small aligned grains and stringers in silicates, cryptocrystalline aggregates and rims around residual titanomagnetite grains. Intermediate stages are probably represented by a common alteration in which ramifying veins of birefringent aggregates cut large titanomagnetite grains and isolate unaltered interiors, but a more complex petrogenesis is indicated by magnetite overgrowing and replacing hematite. In general thermomagnetic determinations show titanomagnetite near pure magnetite in composition to be the remanence carrier (Fig. 5) and a low temperature kink is no longer present in the curves. The hematite has not proved detectable by this method and confirms the conclusion suggested by demagnetization curves (Fig. 2) that titanomagnetite is the predominant remanence carrier.

Interpretation of Palaeomagnetic Poles

The results from the Ballantrae Complex merit close comparison with the Younger Gabbros of Aberdeenshire (Sallomy and Piper 1973), the best-established Lower Palaeozoic result from Britain N of the Highland Boundary Fault. It is known that these rocks were magnetized after the regional folding, and under conditions of protracted cooling the critical range of blocking temperatures would have been lowered, possibly to below 400°C if these temperatures were maintained for periods of the order of 1 m.y. (Pullaiah *et al.* 1975). However, the age of magnetization is certain to lie between the Rb–Sr age 486±17 m.y. (Pankhurst 1970) and K–Ar age, (461±5 m.y.), the latter regarded by Pankhurst as defining the time of cooling through 200°C. Similar arguments apply to the temperature of magnetization of the Ballantrae gabbros and serpentinites, although since these are high-level rocks it is likely that magnetization was acquired close to laboratory-determined blocking temperatures. Here also the K–Ar ages define a minimum possible age for the magnetization, and it is therefore probable that the Ballantrae Complex and Aberdeenshire gabbros were magnetized within a few million years of one another in Lower Ordovician times, an observation consistent with their uniformly-similar polarity. The palaeolatitudes computed from the palaeomagnetic data are: Aberdeenshire gabbros ($I = +34°$, $\lambda = 18\frac{1}{2}°S$), Ballantrae serpentinites ($I = +32°$, $\lambda = 17\frac{1}{2}°S$) and Ballantrae gabbros ($I = +36°$, $\lambda = 20°S$). The palaeolatitudes of the Ballantrae rocks are thus appreciable closer to the Aberdeenshire gabbros and second rank data from Ireland N of the Iapetus suture

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(Connemara gabbros and Mweelrae ignimbrites, Briden *et al.* 1973) than they are to any first-rank data from Britain S of the Southern Uplands. The mean inclination of the four first-rank (defined here as based on more than ten sites and fifty samples) Lower Ordovician (Llanvirn/Llandeilo) palaeomagnetic results from Britain S of the Southern Uplands Fault is 46° equivalent to a palaeolatitude of 27°. These results are derived from the Builth Volcanic Series, central Wales, eastern and western outcrops of the Borrowdale Volcanic Group, and the Eycott Group of the northern Lake District, and are summarized by Briden *et al.* 1973 and Faller *et al.* 1977.

A possible reconstruction of Britain N and S of the Iapetus suture is shown in Figure 6. It is important to remember that palaeomagnetism provides no control of palaeolongitude, and strike-slip motion across the Iapetus suture may be different from that implied by this figure.

The S palaeomagnetic pole from the Bail Hill Volcanic Series lies slightly further to the S on present day coordinates than other British Ordovician poles (Briden *et al.* 1973, fig. 1), but unfortunately there are few results of Caradocian age with which to compare this pole, and results from the North Wales igneous province (Thomas and Briden 1976) suggest that the geomagnetic field was anomalous for at least part of this interval. The only other results from Caradocian age rocks are derived from six tuffaceous units in the Welsh Borderlands (Piper 1978) yielding a mean direction of D = 178°, I = +26° (α_{95} = 18). These data suggest a palaeolatitude difference between Bail Hill (2±6°S) and Shropshire 13½ ± 12°S) of 11½° compared with the present difference of 3°. Thus there is slight evidence for separation across the Iapetus Ocean in early Caradocian times but in view of the small amount of evidence relating to this point, this must be regarded as a very tentative conclusion only.

Discussion

The palaeomagnetic evidence supports a model such as that of Phillips *et al.* (1976) in which the Southern Uplands region formed the northern margin of an Iapetus Ocean *c* 1000 km wide in Lower Ordovician times (Fig. 6). Powell (1970, 1971) has summarized evidence for a sialic basement to this region and Upton *et al.* (1976) suggest that metamorphic clasts in a Carboniferous vent imply continental crust underlying at least part of the Midland Valley. It is probably correct therefore, to regard these as sialic blocks coupled to the orthotectonic Caledonides by Lower Ordovician (post-Arenig) times. The LISPB seismic profile across northern Britain (Bamford *et al.* 1976) has also confirmed that the only major break in the sub-Palaeozoic basement and the Moho occurs beneath the Southern Uplands, coincident with the Iapetus suture of Figure 6.

A separation of the order implied by the palaeomagnetic data is in accord with the palaeontological evidence for a physical barrier between the Girvan region and areas further to the S. This is reflected in the different graptolite faunas but is most strikingly confirmed by the benthonic faunas and Reed (1903–6) first noted that only about 25

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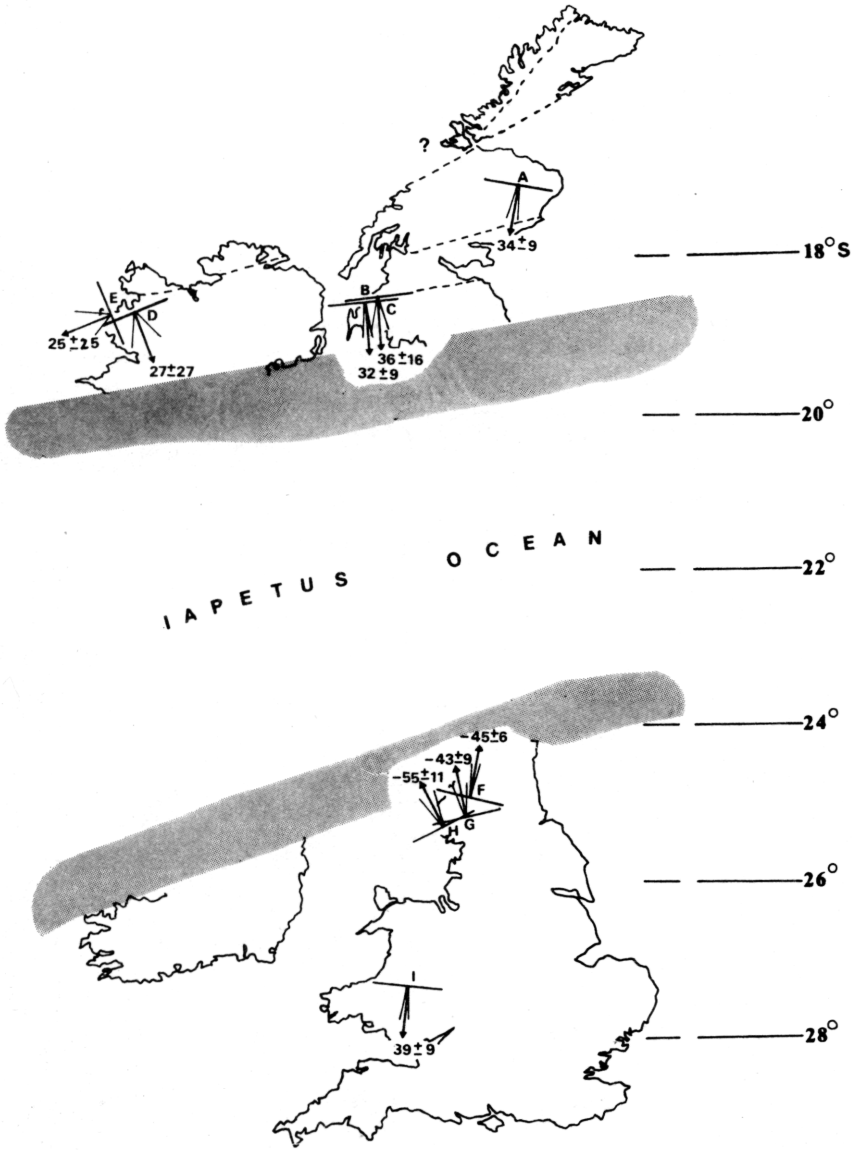


FIG. 6. Reconstruction of the Iapetus (Proto-Atlantic) Ocean in Lower Ordovician (Llanvirn-Llandeilo) times derived from the mean palaeolatitudes determined from palaeomagnetic date. The palaeomagnetic results for Lower Ordovician times are illustrated as mean directions of magnetization with 95% confidence limits, and are derived from A, (Younger) Aberdeenshire Gabbros; B, Ballantrae Serpentinites; C, Ballantrae Gabbros; D, Mweelrae Ignimbrites; E, Connemara Gabbros; F, Eycott Group Lavas; G, Eastern Borrowdale Volcanics; H, Western Borrowdale Volcanics; I, Builth Volcanic Series. Only first-rank data from south of the Iapetus suture are shown; these are supplemented by a considerable body of second rank data summarized by Briden *et al.* (1973) and Piper (1977). The results from Ireland are not used in computation of the mean palaeolatitudes and include one result (E) subject to local tectonic rotation; the magnetic inclinations are however, valid and support a model of crustal separation. The position of Scotland N of the Great Glen Fault is inferred from the climactic metamorphic zonation (Winchester 1973).

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per cent of the trilobite types occuring in the Girvan area occur in England and Wales. This difference is most marked in early Caradoc times and disappears at the beginning of the Silurian period; Williams (1974) notes the ubiquity of brachiopod faunas after mid-Caradocian times. Although this "middle" Ordovician diversity is evidence for a barrier to migration such as a deep ocean basin with turbidite deposition along unstable margins, the disappearance of provincialism in Late Ordovician times cannot necessarily be taken to define the time of final closure of the ocean basin, because the distribution of faunas may relate to other factors such as the contraction of temperature/salinity régimes related to growth of polar ice caps (Skevington 1974). Independent tectonic and magmatic evidence however (Phillips *et al.* 1976; Brown and Hennessey 1977) suggests that closure was complete by Silurian times.

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